

HOMWORK 4

Data Science vs PM2.5

PM2.5 in Chiang Mai: From Raw Data to a Next-Day Warning
Recommendation

DS-270702 Data Science Programming

Chiang Mai University

Individual Homework Report

Study period: 1 January 2023 - 31 August 2026

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1. Problem Background

PM2.5 is fine particulate matter with an aerodynamic diameter of 2.5 micrometres or smaller. In Northern Thailand, high PM2.5 concentrations are a recurring seasonal problem. The assignment context identifies agricultural residue burning, forest fires, transboundary smoke, stable atmospheric conditions, and mountain topography as important contributors to the regional pollution episode. These conditions matter because elevated particulate pollution affects schools, outdoor activities, tourism, and respiratory health.

This study uses Thailand's 24-hour PM2.5 standard of 37.5 ug/m³ as the operational threshold for a dangerous day. The threshold is important for this analysis because the machine-learning task is framed as a decision problem: whether the next day's daily mean PM2.5 will exceed 37.5 ug/m³.

The study focuses on one coordinate in Chiang Mai (18.80 N, 99.00 E) and covers 1 January 2023 through 31 August 2026. Rather than only confirming that the burning season is polluted, the analysis asks when exceedances are most concentrated, which observed weather variables are associated with PM2.5, and whether next-day dangerous conditions can be detected well enough to support an early-warning decision.

The intended audience for the recommendation is provincial decision-makers and organisations that plan outdoor activities, especially schools and groups responsible for communicating risk to sensitive populations. The predictive model is treated as decision support, not as a replacement for official monitoring or public-health advice.

2. Approach

The analysis was designed around three linked questions: (1) How severe and seasonal is the PM2.5 problem at the selected Chiang Mai coordinate? (2) Which same-day weather conditions are associated with high PM2.5? (3) Can information available on day t identify whether day $t+1$ will exceed 37.5 ug/m³?

Before analysis, the expected pattern was that PM2.5 would be concentrated in the dry-season months and that atmospheric moisture or rainfall might be negatively associated with PM2.5. For prediction, the expectation was that today's pollution level would be informative for tomorrow, making a persistence rule a strong baseline. Therefore, any machine-learning model had to be compared directly with persistence rather than judged from accuracy alone.

Because dangerous days are the minority class, recall is especially important. A false negative means a dangerous day is missed. The analysis therefore evaluates accuracy, precision, recall, and F1-score and explicitly examines false-negative cases.

3. Method

3.1 Data sources and acquisition

Two Open-Meteo endpoints were requested programmatically with timezone set to Asia/Bangkok. Raw API responses were saved before transformation. The air-quality endpoint supplied PM2.5, PM10, carbon monoxide, and dust. The historical weather endpoint supplied temperature at 2 m, relative humidity at 2 m, wind speed at 10 m, wind direction at 10 m, precipitation, and surface pressure.

Source	Endpoint / content	Period / rows
Open-Meteo Air Quality API	air-quality-api.open-meteo.com/v1/air-quality; hourly PM2.5, PM10, CO, dust	2023-01-01 to 2026-08-31; 32,136 hourly rows
Open-Meteo Historical Weather API	archive-api.open-meteo.com/v1/archive; hourly weather variables	2023-01-01 to 2026-08-31; 32,136 hourly rows

The two hourly time axes were tested and were identical. An inner join on time produced 32,136 rows, so no hourly observation was lost during the join. The joined hourly data were then aggregated to Thailand local calendar days, producing 1,339 daily observations.

3.2 Data quality and daily aggregation

Missing-value percentages were calculated for every joined variable and all reported 0.00% missing. Duplicate-row checks also returned zero duplicates. The absence of missing values should not be interpreted as proof that a ground sensor recorded every hour: the Open-Meteo products used here are gridded/model-based environmental data, so complete coverage is plausible because of the upstream data product rather than perfect local sensor availability.

Daily PM2.5, PM10, carbon monoxide, dust, temperature, relative humidity, wind speed, wind direction, and surface pressure were calculated as daily means; precipitation was summed by day. Daily aggregation was performed only after requesting Asia/Bangkok timestamps, so the daily boundaries correspond to local Thailand dates.

3.3 Target, features, and split

The classification target is `danger_tomorrow` = 1 when tomorrow's daily mean PM2.5 is greater than 37.5 ug/m3, otherwise 0. The final day was removed because it has no next-day outcome. This created 1,338 modeling rows: 1,178 safe outcomes (88.04%) and 160 dangerous outcomes (11.96%).

Ten same-day features were used: PM2.5, PM10, carbon monoxide, dust, temperature, relative humidity, wind speed, wind direction, precipitation, and surface pressure. A leakage audit confirmed that `pm25_tomorrow` and `danger_tomorrow` were not included among predictors.

The split was chronological: the first 80% (1,070 rows, 2023-01-01 to 2025-12-05) formed the training set and the final 20% (268 rows, 2025-12-06 to 2026-08-30) formed the test set. This is appropriate for a next-day forecasting task because the evaluation simulates prediction on observations that occur after the training period.

3.4 Models and evaluation

The persistence baseline predicts tomorrow's dangerous/safe status using today's PM2.5 status. Two machine-learning models were then fitted: balanced Logistic Regression with standardized predictors, and a balanced Random Forest with 300 trees, maximum depth 8, minimum leaf size 3, and `random_state` 42. Performance was evaluated on the same chronological test set using accuracy, precision, recall, and F1-score.

The submitted notebook does not contain a TimeSeriesSplit cross-validation result. Therefore, this report does not claim a cross-validation score; the reported model comparison is based on the held-out chronological test period.

4. Results

4.1 PM2.5 severity and seasonality

Across 1,339 daily observations, mean PM2.5 was 23.81 $\mu\text{g}/\text{m}^3$ and the median was 21.25 $\mu\text{g}/\text{m}^3$. The minimum daily mean was 5.86 $\mu\text{g}/\text{m}^3$ and the maximum was 116.70 $\mu\text{g}/\text{m}^3$. In total, 160 days (11.95%) exceeded 37.5 $\mu\text{g}/\text{m}^3$.

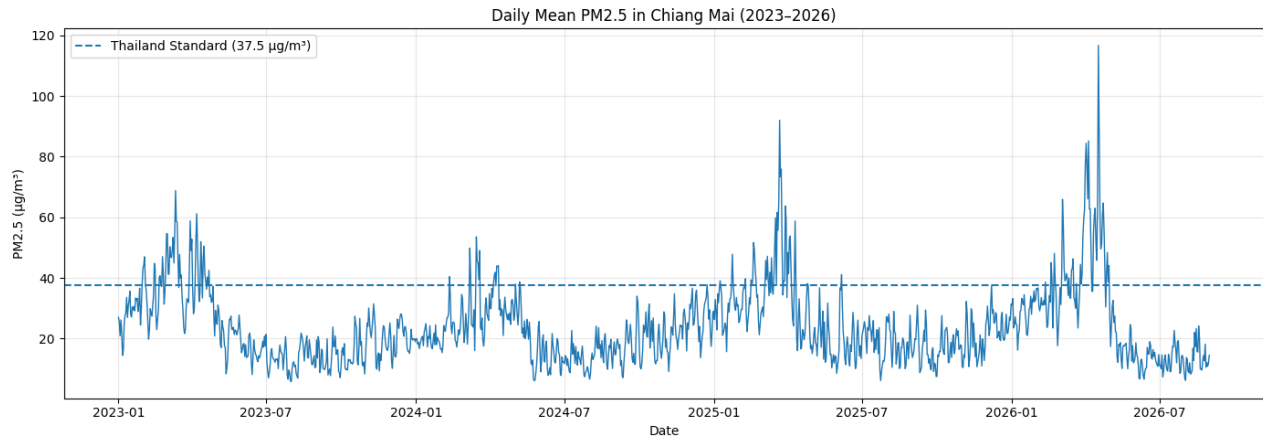


Figure 1. Daily mean PM2.5 at the Chiang Mai study coordinate, with the 37.5 $\mu\text{g}/\text{m}^3$ threshold.

Figure 1 shows that exceedances are not distributed evenly through time. Peaks recur in the early part of the year, supporting a clear seasonal pattern rather than a constant year-round problem.

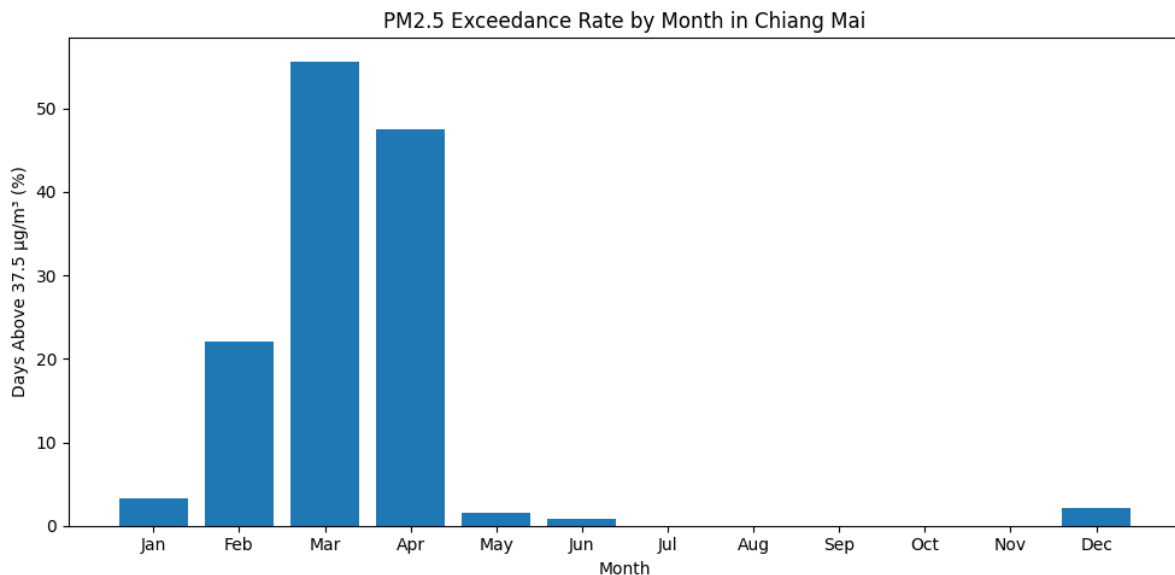


Figure 2. Monthly percentage of observed days exceeding 37.5 $\mu\text{g}/\text{m}^3$.

March had the highest exceedance rate at 55.65%, followed by April at 47.50% and February at 22.12%. July through November had no exceedance days in the analyzed data. This identifies February-April, especially March and April, as the period in which preparedness has the strongest empirical justification.

4.2 Weather associations

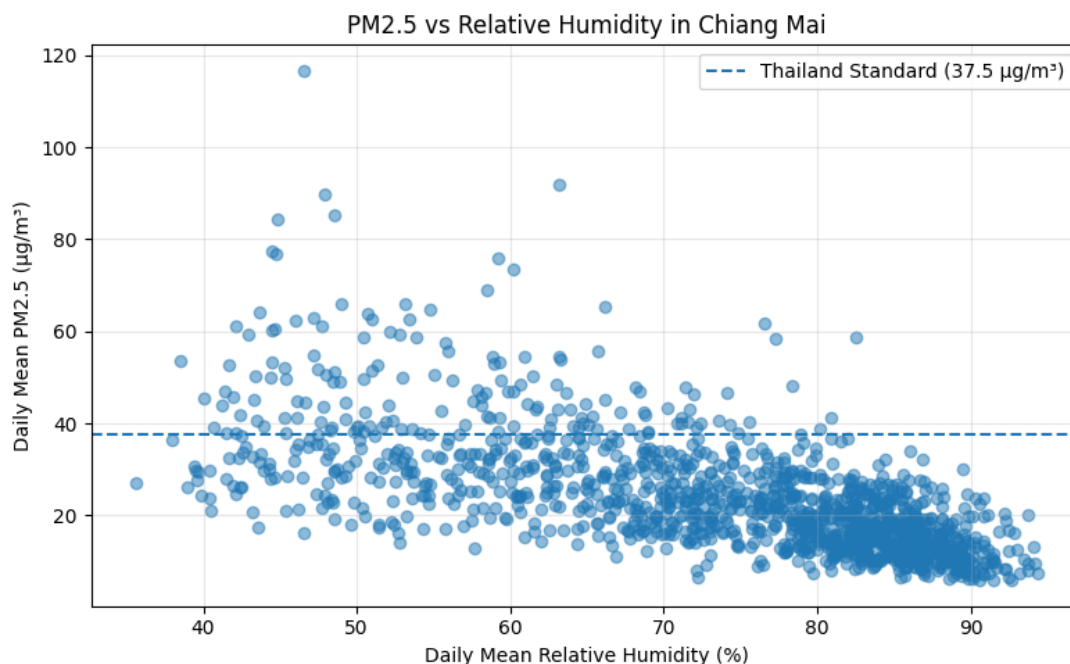


Figure 3. Daily PM2.5 versus relative humidity.

Relative humidity had the strongest linear association among the selected weather variables ($r = -0.657$). Precipitation was also negatively correlated with PM2.5 ($r = -0.450$). Wind speed was almost uncorrelated ($r = 0.025$), while temperature ($r = 0.118$) and surface pressure ($r = 0.217$) had weak positive correlations. These are associations, not causal estimates.

4.3 Next-day model performance

Model	Accuracy	Precision	Recall	F1
Persistence Baseline	0.918	0.788	0.788	0.788
Logistic Regression	0.843	0.560	0.904	0.691
Random Forest	0.899	0.719	0.788	0.752

The persistence baseline achieved the best F1-score (0.788) and the highest overall accuracy (0.918). Random Forest did not beat the baseline on F1 and matched its recall at 0.788. Logistic Regression had a lower F1-score (0.691) but the highest recall (0.904). Thus, the machine-learning models did not improve the balanced F1 result over persistence; Logistic Regression is only preferable when the operational priority is to miss as few dangerous days as possible and additional false alarms are acceptable.



Figure 4. Logistic Regression confusion matrix on the chronological test set.

The Logistic Regression model correctly detected 47 of 52 dangerous test days and missed 5. It produced 37 false-positive warnings. The five false negatives occurred when PM2.5 rose from a currently safe level to cross the threshold on the next day; the increases were approximately 5.55, 7.50, 15.50, 5.77, and 11.14 ug/m3. Sudden threshold-crossing increases are therefore a specific failure mode of the current feature set.

4.4 Required checkpoints

Checkpoint	Evidence / answer
C1 - Time	Both endpoints used timezone Asia/Bangkok; their hourly time axes were identical. Daily aggregation therefore used Thailand local calendar days.
C2 - Join	Air quality: 32,136 rows; weather: 32,136 rows; inner join: 32,136 rows. No rows were lost.
C3 - Missing values	Every hourly and daily column reported 0.00% missing; duplicates = 0. Completeness is consistent with gridded/model-based Open-Meteo products rather than proof of perfect local sensor coverage.
C4 - Comparing places	Not applicable to spatial comparison: this analysis uses one Chiang Mai coordinate only and makes no claim that one location is cleaner or dirtier than another.
C5 - Model vs baseline	Baseline F1 = 0.788. Logistic Regression F1 = 0.691; Random Forest F1 = 0.752. Neither model beat the baseline F1 on the same test set. Logistic Regression had higher recall (0.904).

C6 - Ground truth	Completed as a supplementary ground-truth check for 31 Aug 2026. Air4Thai hourly PM2.5 at the Yupparaj Wittayalai School station, Si Phum, Mueang Chiang Mai, was digitised from the official historical chart and averaged to approximately 8.80 ug/m3. The Open-Meteo daily mean at 18.80 N, 99.00 E was 11.27 ug/m3, giving an absolute difference of approximately 2.46 ug/m3. The values are reasonably close but are not expected to match exactly because Air4Thai is a ground-station measurement whereas Open-Meteo is a gridded/model-based product, and the sites/methods are not identical.
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4.5 Results that did not work out

Wind speed was initially examined as a possible explanatory weather variable, but its correlation with PM2.5 was only 0.025, so it did not provide a useful simple linear relationship. More importantly, neither Logistic Regression nor Random Forest beat the persistence baseline on F1-score. These negative results are retained because they constrain what can reasonably be recommended.

5. Conclusion

The analysis shows a strong seasonal PM2.5 problem at the selected Chiang Mai coordinate. The most concentrated exceedance risk occurred in February-April, with March (55.65%) and April (47.50%) standing out. Lower relative humidity and lower precipitation were associated with higher PM2.5, although the analysis cannot establish that these weather variables cause the pollution episodes.

For next-day warning, persistence is a strong benchmark and achieved the best F1-score. Logistic Regression did not outperform it overall, but its recall of 0.904 means it detected 47 of 52 dangerous test days. This may still be useful where the cost of a missed hazardous day is judged higher than the cost of a false alarm.

Recommendation. Provincial agencies and schools in Chiang Mai should strengthen PM2.5 preparedness and communication during February-April, particularly March and April. If a next-day model is used as an additional decision-support signal, a high-recall configuration such as the tested Logistic Regression can be considered for early precautionary alerts, but official monitoring and public-health advisories should remain authoritative. The additional false positives must be communicated as a known trade-off.

The evidence does not support replacing official monitoring, claiming causal effects from weather, or generalising this single-coordinate result to the whole province. Future work should add fire-hotspot or biomass-burning indicators, boundary-layer information, forecast variables available at prediction time, multiple locations, and a larger set of paired Air4Thai/Open-Meteo observations. The supplementary Air4Thai check on 31 Aug 2026 showed an approximate 2.46 ug/m3 difference, but one date is not sufficient to establish measurement accuracy.

5.1 Limitations

- Open-Meteo values are gridded/model-based and may differ from a specific ground monitoring station.
- Only one Chiang Mai coordinate was analyzed, so spatial variation across the province was not tested.
- The study ends in August 2026; 2026 is an incomplete calendar year.
- Weather correlations are associative and do not establish causation.

- Same-day observations do not fully capture sudden next-day PM2.5 increases.
- The high-recall Logistic Regression produced 37 false-positive warnings in the test period.
- Fire hotspots, agricultural burning, traffic emissions, and transboundary pollution were not directly included.
- A supplementary Air4Thai ground-truth check was completed for 31 Aug 2026; however, the Air4Thai daily mean was estimated by digitising the official hourly chart, so the comparison should be treated as approximate.
- The notebook reports a chronological hold-out test but does not report TimeSeriesSplit cross-validation.

References

Open-Meteo. Air Quality API. Data endpoint used by the submitted fetch script.

Open-Meteo. Historical Weather API. Archive endpoint used by the submitted fetch script.

Pollution Control Department, Thailand. Air4Thai air-quality monitoring service; historical hourly PM2.5 chart for Yupparaj Wittayalai School station, Chiang Mai, 31 Aug 2026.

DS-270702 Data Science Programming. Homework 4: PM2.5 in Northern Thailand lab sheet.

Appendix A. Traceability of Key Outputs

Repository output	Purpose
data/raw/open_meteo_air_quality.json	Raw air-quality API response
data/raw/open_meteo_weather_archive.json	Raw historical weather API response
data/processed/hourly_merged.csv	Joined hourly dataset
data/processed/daily_merged.csv	Daily analysis dataset
data/processed/modeling_dataset.csv	Modeling dataset
outputs/results/model_comparison.csv	Model metrics
outputs/results/false_negative_cases.csv	Missed dangerous-day cases
outputs/figures/fig01_pm25_timeseries.png	Time-series figure
outputs/figures/fig02_pm25_exceedance_rate.png	Monthly exceedance-rate figure
outputs/figures/fig03_pm25_humidity.png	Humidity association figure
outputs/figures/fig04_confusion_matrix.png	Logistic Regression confusion matrix

AI Tool Disclosure

Complete and edit this table so that it accurately describes your own use of AI tools. The assignment requires this disclosure as the final page of the report.

Where you used it	What you asked for	What you changed or rejected
Coding / debugging	Help explain, debug, and organise Python analysis steps.	Reviewed the generated suggestions and kept only code/results I could explain and verify.
Report drafting	Help organise notebook results into the required report structure.	Checked all numerical claims against notebook outputs; unsupported claims were removed or marked pending.
Language / presentation	Help improve English clarity and figure interpretation.	Edited wording to match the evidence and retained limitations and negative results.

Student confirmation: I am responsible for every line of code, analysis, and text submitted and can explain the choices made in this work.